Course:- Theory of Computations

Name:- G. Venkatesh Reddy
Reg No:- 192372262
Course Code:- CSA 1303

CO1 - AT2

1) NFA for Automatic Door System.

A) Alphabet : $\Sigma = \{P, N\}$

P = person detected

N = no person detected

States:-

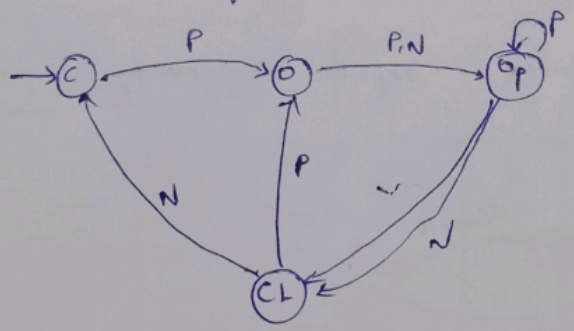
C = closed (start)

O = opening

Op = open

CL = Closing

State Diagram:-



Initial states:-

C (start state)

State Transitions:-

C = closed (start)

O = opening

Op = open

C → P → O

O → P,N → Op

Op → P → Op (door remains open)

Op → N → {Op, CL}

CL → N → C

CL → P → O

Transition table:-

State	P	N
C	O	C
O	Op	Op
Op	Op	{Op, CL}
CL	O	C

2) Finite Automation for Traffic light control.

1) Alphabet: Σ = {timer R, timer G, timer Y, pedestrian, emergency}

States: R = Red (start)

G = Green

Y = Yellow

Normal timed sequence:

$$R \xrightarrow{\text{timer R}} G \xrightarrow{\text{timer G}} Y \xrightarrow{\text{timer Y}} R$$

Timer durations (example values)

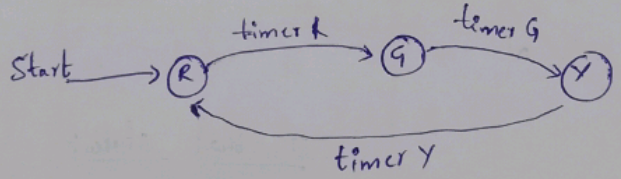
Red: 60s

Green: 40s

Yellow: 5s

The sequence R → G → Y → R repeats continuously.

Three state cycle:



Transition table (normal cycle):

State	next state
R	G
G	Y
Y	R

3) DFA accepting strings ending with ab

A) Alphabet = $\Sigma = \{a, b\}$

Given string w = aaaabab

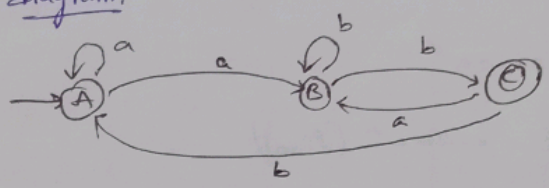
States:-

A → start state

B → last symbol is a

C → string ends with ab (final state).

State diagram:-



Transition Table:-

State	a	b
A	B	A
B	B	C
C	B	A

Final State:-

C (accepting)

∴ The string aaaabab is accepted.

4) DFA accepting only c, bc, bcaaa.

A) Alphabet: $\Sigma = \{a, b, c\}$

States:

S – start state

C – accepts "c" (final)

B – prefix "b"

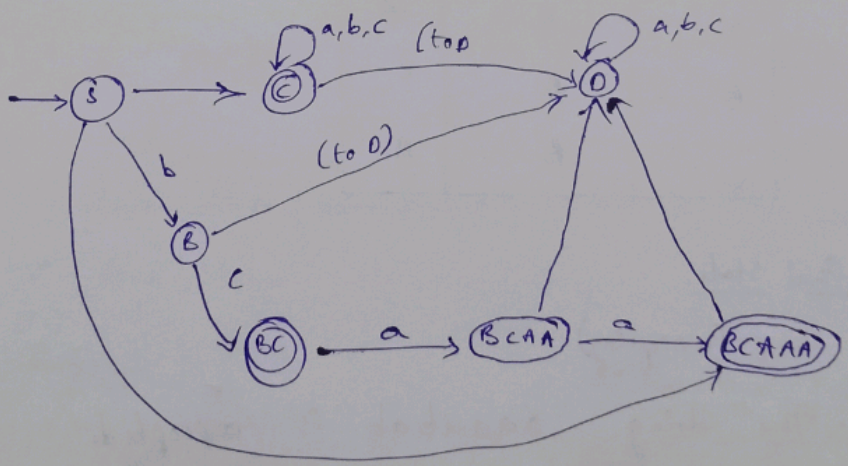
BC – accepts "bc" (final)

BCA – prefix "bc a"

BCAA – prefix "bc aa"

BCAAA – accepts "bcaaa" (final)

D – dead state (trap)



Transition table:

State	a	b	c
S	D	B	C
C	D	D	D
B	D	D	BC
BC	BCA	D	D
BCA	BCAA	D	D
BCAA	BCAAA	D	D
BCAAA	D	D	D
D	D	D	D

5) DFA accepting all non-empty strings over {a,b}

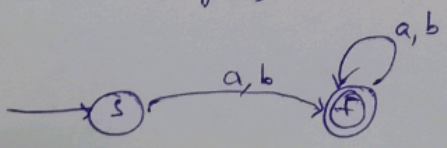
A) Alphabet = Σ = {a,b}

States:

S – start state (non-final)

f – final state

State diagram:



Transition Table:

State	a	b
S	f	f
f	f	f